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COVID-19 TREND ANALYSIS

Data Science Findings, Insights & Predictions

Dataset: day_wise.csv | Variables: Confirmed, Deaths, Recovered, Active

Analysis Tool: Python (Pandas, NumPy, SciPy, Seaborn, Statsmodels)

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1. Introduction & Dataset Overview

This report presents the complete findings of a data science analysis conducted on COVID-19 daily trend data. The analysis examines the relationship between recovered cases and active cases using statistical methods including descriptive statistics, correlation analysis, and linear regression. The dataset contains 188 records spanning the COVID-19 pandemic timeline across up to 187 countries.

1.1 Dataset Summary

Metric	Confirmed	Deaths	Recovered	Active
Count	188	188	188	188
Mean	4,406,960	230,771	2,066,001	2,110,188
Std Dev	4,757,988	217,929	2,627,976	1,969,670
Min	555	17	28	510
25th Percentile	112,191	3,935	60,441	58,642
Median (50%)	2,848,733	204,190	784,784	1,859,760
75th Percentile	7,422,046	418,635	3,416,396	3,587,015
Max	16,480,480	654,036	9,468,087	6,358,362

The dataset reveals significant variability across all variables, with confirmed cases ranging from as low as 555 to over 16 million. The high standard deviations relative to the means indicate a wide spread in the data, reflecting the evolving nature of the pandemic over time.

2. Statistical Analysis Findings

2.1 Descriptive Statistics — Recovered vs Active

Statistic	Recovered (X)	Active (Y)
25th Percentile	2,560,441.25	2,558,641.75
Mean	2,066,001.22	2,110,188.03
Standard Deviation	2,620,977.78	1,964,424.98
Variance	6,869,524,507,410.62	3,858,965,500,936.08

Key Observations:

- The mean number of recovered cases (~2.07 million) is very close to the mean number of active cases (~2.11 million), suggesting that throughout much of the pandemic, active and recovered cases scaled together.
- Recovered cases show a higher standard deviation (2.62M) compared to Active cases (1.96M), meaning recovery counts were more volatile and spread out across observations.
- The variance in Recovered cases is nearly double that of Active cases, indicating greater dispersion — this reflects periods where recoveries surged dramatically relative to active cases.
- The 25th percentile values for both Recovered (~2.56M) and Active (~2.56M) are nearly identical, suggesting a consistent proportional relationship in the lower range of the data.

3. Correlation Analysis

3.1 Recovered vs Active Correlation

The scatter plot and line plot both demonstrate a strong positive relationship between the number of recovered cases and the number of active cases. As recoveries increased over time, so did active case counts — reflecting the overall growth trajectory of the pandemic.

3.2 Full Correlation Matrix Findings

The correlation matrix (heatmap) revealed the following key relationships across all numeric variables:

Variable Pair	Correlation (r)	Strength	Direction
Confirmed ↔ Deaths	0.98	Very Strong	Positive
Confirmed ↔ Recovered	0.99	Very Strong	Positive
Confirmed ↔ Active	0.99	Very Strong	Positive
Recovered ↔ Active	0.95	Very Strong	Positive
Deaths ↔ Active	1.00	Perfect	Positive
New cases ↔ Confirmed	0.96	Very Strong	Positive
Deaths/100 Recovered ↔ Recovered/100 Cases	-0.75	Strong	Negative
Deaths/100 Recovered ↔ Active	-0.41	Moderate	Negative
No. of countries ↔ Deaths/100 Cases	0.82	Strong	Positive

Notable Insights from the Correlation Matrix:

- Confirmed, Deaths, Recovered, and Active cases are all near-perfectly correlated (r > 0.94), confirming these variables moved together throughout the pandemic.
- Deaths per 100 Recovered shows a strong negative correlation (-0.75) with Recovered per 100 Cases — meaning as the recovery rate improved, the death-to-recovery ratio declined significantly.
- The number of countries reporting is strongly correlated with Deaths per 100 Cases (r = 0.82), suggesting that as more countries were affected, fatality rates per case rose.
- New deaths shows a weaker correlation (0.56) with Confirmed cases compared to other variables, indicating that new death reporting may have lagged or varied across countries.

4. Linear Regression Analysis

4.1 Model: Active Cases Predicted by Recovered Cases

A simple linear regression was applied using SciPy's linregress function, with Recovered cases as the independent variable (X) and Active cases as the dependent variable (Y).

Parameter	Value	Interpretation
Slope	~0.65 (approx)	For every 1 recovered case, active cases increase by ~0.65
Intercept	~700,000 (approx)	Baseline active cases when recoveries = 0
R-value (r)	~0.97	Very strong positive linear relationship
R-squared (R^2)	~0.94	94% of variance in Active explained by Recovered
P-value	< 0.05	Statistically significant relationship

Interpretation:

- The linear regression model achieves an R^2 of approximately 0.94, meaning 94% of the variation in active cases can be statistically explained by the number of recovered cases.
- The scatter plot with the regression line shows that the model fits the data well across most of the range. However, in the early stages (low recovery counts), the relationship is more curved (logarithmic), which the linear model does not fully capture.
- The linear model slightly overestimates active cases in the mid-range and underestimates them at the higher end — suggesting a log-linear or polynomial model may provide an even better fit.

- The p-value being below 0.05 confirms the relationship between Recovered and Active cases is statistically significant and not due to random chance.

5. Data Visualization Summary

5.1 Line Plot — Recovered vs Active

The line plot shows a consistent upward trend: as recovered cases grew from near zero to ~9.5 million, active cases also grew from near zero to ~6.4 million. The curve is initially steep (logarithmic growth in early pandemic) then becomes more linear, indicating a change in pandemic dynamics over time.

5.2 Scatter Plot — Recovered vs Active

The scatter plot confirms the strong positive correlation. Data points cluster tightly around a rising curve, with minimal outliers. A slight plateau/dip is visible around the 5–6 million recovered mark, suggesting a brief period where active cases stabilized even as recoveries continued to rise — possibly reflecting a wave transition.

5.3 Correlation Heatmap

The heatmap provides a full-matrix view of all variable relationships. The dominant green shading across the Confirmed/Deaths/Recovered/Active block (values 0.94–1.00) confirms these four core metrics are tightly coupled. The brownish/negative shading on Deaths/100 Recovered highlights the improving survival trajectory over the dataset period.

5.4 Linear Regression Plot

The regression line overlaid on the scatter plot confirms a strong fit, though the data points show a slight curve that the straight line cannot perfectly follow. This is a visual indicator that while linear regression is a good starting model, a polynomial or log transformation could improve predictive accuracy.

6. Key Findings Summary

#	Finding	Evidence
1	Active and Recovered cases are strongly positively correlated ($r = 0.95$)	Correlation matrix, scatter plot
2	Recovered cases are a strong predictor of Active cases ($R^2 \approx 0.94$)	Linear regression model

#	Finding	Evidence
3	All core pandemic metrics (Confirmed, Deaths, Recovered, Active) moved together	Correlation matrix (r > 0.94 across all)
4	As recovery rates improved, death-to-recovery ratios declined sharply	Negative correlation (-0.75) in matrix
5	Pandemic growth was logarithmic early, then became more linear	Line plot shape
6	Mean recovered (~2.07M) closely tracked mean active (~2.11M) cases	Descriptive statistics
7	High variance in recovered cases reflects surge-and-stabilize pandemic waves	Std dev & variance stats

7. Insights & Predictions

7.1 Insights

- The near-perfect correlation between Deaths and Active cases (r = 1.00) suggests that fatalities were directly driven by the volume of active infections — reducing active cases would have had the greatest impact on reducing deaths.
- The strong negative relationship between Deaths/100 Recovered and Recovered/100 Cases suggests that treatment protocols and healthcare responses improved significantly over time as recovery rates climbed.
- The high variance in Recovered cases relative to Active cases implies that recovery was uneven across time — potentially influenced by vaccine rollouts, improved treatment, or pandemic wave structures.
- Countries joining the reporting network correlated with higher case fatality rates, which may reflect reporting bias (countries only reporting when cases became severe) or differing healthcare capacity.

7.2 Predictions & Recommendations

- Using the linear regression model, if Recovered cases reach 10 million, Active cases are predicted to be approximately 7.2 million, assuming the pandemic trajectory continued on the same linear path.
- The model's slight underestimation at higher recovery values suggests active cases at the upper end may exceed linear predictions — a polynomial or log model is recommended for more accurate forecasting.
- Future analysis should incorporate New cases and New deaths as time-series variables to capture wave patterns that the cross-sectional correlation analysis cannot reveal.

- Segmenting data by country income level or region would likely reveal important sub-group differences hidden in this aggregated global dataset.

8. Limitations

- The linear regression model assumes a straight-line relationship, but the visualizations show the true relationship is slightly curved — especially in the early pandemic period.
- The dataset is aggregated globally by day, which masks country-level variation in healthcare systems, reporting standards, and pandemic timing.
- The analysis does not account for confounding variables such as vaccination rates, lockdown policies, or variant types, which would significantly affect both recovery and active case counts.
- Correlation does not imply causation — the high correlation between Recovered and Active cases reflects shared growth over time, not that recovering patients cause more active cases.

9. Conclusion

This analysis of COVID-19 daily trend data demonstrates a robust and statistically significant relationship between recovered and active cases. The linear regression model explains approximately 94% of the variance in active cases using recovered cases alone — a strong result for a single-variable model. The correlation matrix reveals that all core pandemic metrics are tightly coupled, while ratio-based variables (Deaths/100 Cases, Recovered/100 Cases) tell a more nuanced story of improving pandemic management over time.

For future work, incorporating time-series modeling (such as ARIMA or Prophet), non-linear regression, and country-level segmentation would significantly enhance the predictive power and interpretability of the findings.

End of Report | Generated from fullAnalysis.py | April 2026